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From Data to Action: Leveraging Dynamic Operating Envelopes to Optimize the Production of Wells

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Abstract

With the widespread adoption of digital technologies in the Oil & Gas industry, a range of solutions have emerged to enhance monitoring and deliver actionable insights. While these tools have improved decision-making and efficiency, their growing complexity has made implementation, integration, and scalability increasingly challenging across different operational environments. This paper introduces a data-driven approach to production optimization through the deployment of dynamic well operating envelopes. The objective of this approach is to convert real-time data into operational actions that improve efficiency and minimize downtime. A key innovation lies in the integration of contextual awareness - achieved by using AI to interpret sensor data alongside operational logs. This enables the system to incorporate past events and production constraints into its decision logic, ensuring recommendations are both accurate and field relevant. This methodology integrates data from multiple sources related to well performance - including subsurface measurements, integrity parameters, and production logs - to train machine learning models that associate flow rates with their corresponding operating envelopes and events. These envelopes are then continuously updated using real-time field data, enabling the system to adapt to evolving operating conditions and recommend optimized operational guidelines. The system is structured around a three-step approach: 1. Integration: Aggregates dispersed data sources into a unified digital platform, enabling contextualization of data from individual wells to entire assets. 2. Analysis: Applies advanced analytics and machine learning to identify patterns, detect deviations, and propose optimized guidelines. 3. Recommendation: Generates automated, context-aware recommendations via dynamic operating envelopes, supporting cross-functional collaboration and enhanced asset understanding. The approach was implemented in a mature offshore asset with aging infrastructure and increasing operational constraints. In its initial deployment, the system identified suboptimal operating zones - such as deviations in wellhead pressure and inefficient gas lift configurations - and recommended corrective actions, which were validated and implemented by field engineers. The adaptive nature of the envelopes allowed for real-time tuning in response to evolving reservoir and surface conditions. This responsiveness led to a direct improvement in production efficiency, resulting in a cumulative gain of over 150 kbbbl in additional production. The integration of multi-source data - including sensors, logbooks, and operational setpoints - was critical in capturing complex system interactions and enabling prescriptive guidance. These results

confirm the potential of dynamic operating envelopes to serve as a foundational layer for autonomous well production optimisation.

Introduction

In recent years, the world has gone through an intense digital transformation that has fundamentally reshaped the way companies compete and operate. The Oil & Gas industry has been a prominent participant in this change, implementing innovative technologies to refine decision-making processes and improve operational efficiency. Tools based on advanced analytics, predictive modeling, and process automation have become widespread, promising to optimize performance across the whole value chain. However, the adoption of these technologies has not been without challenges. As digital platforms have evolved rapidly, they have also introduced significant complexity, particularly when integrated into existing systems. Many companies rely on older legacy infrastructures that often operate in isolation, making it difficult to seamlessly incorporate new tools. This creates a fragmented technological landscape, where disconnected systems and a lack of standardization hamper the scalability and consistency of solutions.

In addition to technological obstacles, organizations within the Oil & Gas sector often encounter operational limitations. Data, one of the most valuable resources in a digital environment, frequently remains siloed in separate systems or locations. Without centralized access to reliable and comprehensive datasets, building actionable insights becomes more difficult. The disparity in workflows and processes between different operational teams only exacerbates the problem, and the workforce is frequently expected to adapt to new and unfamiliar systems. This leads to steep learning curves that slow the adoption of digital tools and reduce their overall effectiveness, as users may struggle to maximize functionality or integrate these solutions into their daily activities.

Despite these obstacles, digital technologies hold significant potential for transformation of daily operations. The key to overcoming these barriers lies in turning the vast amounts of collected data into clear, practical insights that drive real actions in the field. This approach not only enhances productivity but also helps companies build more adaptable and sustainable business models that can better respond to future challenges.

Operating Envelopes

The concept of Well Operating Envelopes (WOE) has been an integral part of the Oil & Gas industry for years, serving as a structured approach to defining the safest and optimal operating ranges for key well parameters, including pressure, temperature, and flow rates. These envelopes incorporate boundaries to ensure operational safety while maximizing production efficiencies, and safeguarding equipment from damage. By setting these limits, operators can mitigate risks related to equipment failure, reservoir damage, or safety incidents, contributing to a more consistent and reliable production performance.

Generating WOE typically involves teamwork across various disciplines, bringing together reservoir engineers, field operations specialists, and other key personnel to ensure a comprehensive approach. These teams rely on tools like nodal analysis to simulate well performance and use engineering calculations to assess how the well behaves under different operating conditions. Sensitivity analyses are often used to explore the impact of variations in parameters and to define acceptable limits. While this process provides valuable insights, creating and maintaining a traditional envelope is often a time-consuming and labour-intensive task. Large amounts of data must be gathered, validated, and analysed, which often delays its final development. As a result, once created these envelopes are rarely revisited, making them static representations that do not account for the dynamic nature of well and reservoir conditions. This creates inefficiencies, as operators miss opportunities to adjust and optimize performance in response to changing conditions.

One possible way to overcome this limitation lies in the use of machine learning algorithms to calculate dynamic operating envelopes. Unlike traditional methods, dynamic WOE's are continuously updated based on real-time field data, allowing them to adapt to changing operational conditions. This approach not only provides a more accurate and up-to-date view of operating boundaries but also drastically reduces the need for manual data preparation and calculations, minimizing the risk of human error. When properly implemented, dynamic operating envelopes enable operations teams to quickly identify deviations from optimal performance and proactively identify improvement opportunities.

Methodology

A key aspect of this methodology involves the use of virtual flow meter (VFM) models to accurately define and manage the operating envelopes for individual wells. VFMs are advanced software tools specifically designed to estimate real-time flow rates without requiring physical flow meters, which can often be expensive or impractical to install. These models analyze extensive datasets from historical well tests and apply machine learning (ML) techniques to uncover patterns and insights. Through this process, VFMs can identify and understand the intricate and often non-linear relationships between various well parameters like wellhead pressure (WHP), wellhead temperature (WHT), and choke opening and the actual flow rates recorded during prior operations (see Figure 1):

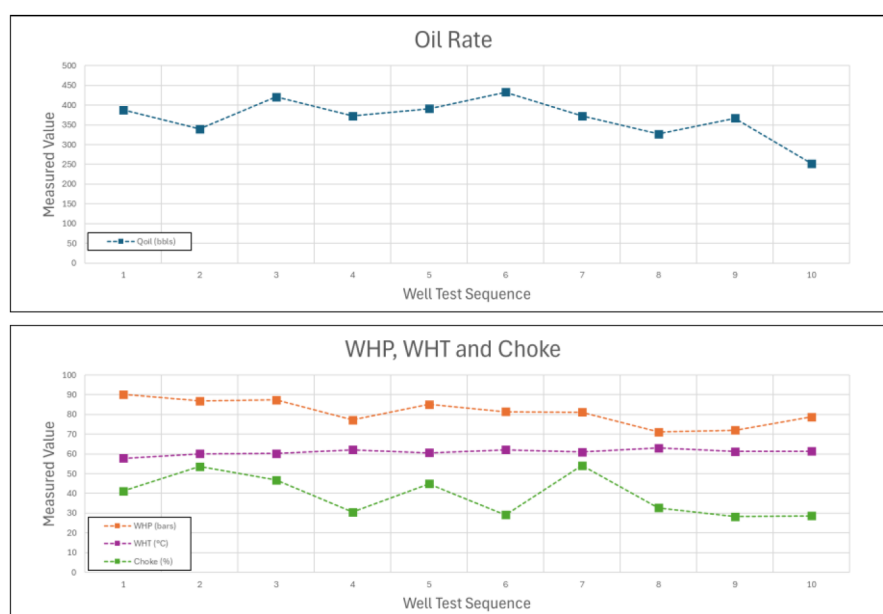


Figure 1—Historical well test data showing actual measured oil rates and the corresponding well parameters (WHP, WHT and Choke).

Unlike traditional approaches reliant on static models or manual calculations, VFMs offer a flexible and responsive way to monitor well performance. Once implemented, these models operate by continuously analyzing data collected from field sensors to deliver up-to-date estimates of production rates for oil, gas, and water (see Figure 2). Sensors installed on wells measure real-time parameters and send the data to the virtual flow meter, which uses its underlying analytical models to calculate flow rates in real-time. The system's ability to provide continuous estimates completely eliminates the need for frequent manual well testing, offering operators a clearer and more immediate picture of ongoing production performance.

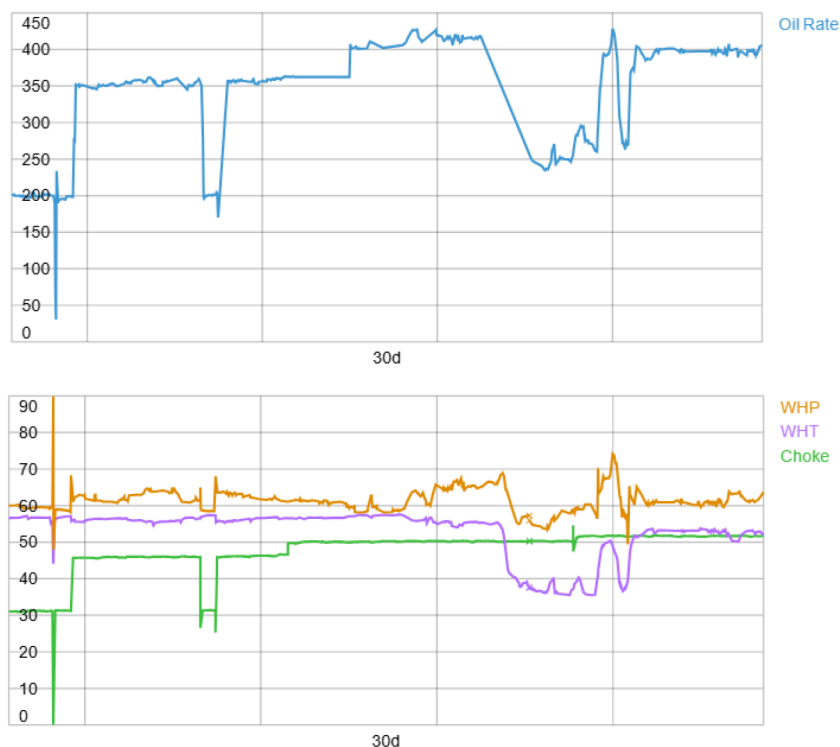


Figure 2—Real VFM-predicted oil rates over a 30-day period derived from corresponding WHP, WHT and Choke parameters.

In addition to estimating flow rates, VFM models can be used backward to determine the optimal well parameters necessary to achieve a desired production target. This process involves simulating various scenarios by adjusting controllable variables and observing how these changes influence the predicted production rates. By automatically running a series of sensitivity analyses (see Table 1), the model can identify the combinations of well parameters that most closely match the targeted output rates.

Table 1—Simulated well parameter ranges (WHP, WHT and Choke) for backward VFM sensitivity analysis with the optimal operational envelope highlighted in gray.

WHP (bars)	WHT (°C)	Choke (%)	Predicted Oil Rate (bbls)	Target Oil Rate (bbls)	Optimal Envelope
70	56	30	344	450	N
70	56	40	339	450	N
70	56	50	453	450	YES
80	58	30	393	450	N
80	58	40	597	450	N
80	58	50	498	450	N
...	...	...	...	...	...
90	60	30	473	450	N
90	60	40	325	450	N
90	60	50	352	450	N
100	62	30	542	450	N
100	62	40	407	450	N
100	62	50	387	450	N

Data driven VFM workflow

The above-described VFM methodology is implemented through a structured workflow consisting of three main steps: selection and preparation of training data, retraining of model to incorporate new information, and inference of flow rate estimations.

Training Dataset. The automated data preparation pipeline runs continuously, ensuring that all models consistently incorporate any new validated data as soon as it becomes available. This whole process starts by identifying the list of well parameters to be used for model training, based on the currently available sensors at well level and the corresponding parameters present in well tests validated by operational teams. A rigorous acceptance criterion is then applied: only the well parameters available across at least 90% of well test data are included in the training dataset, with the resulting curated list of well parameters being referred to as the *Well_parameters_list*.

Model Retraining. The decision to retrain models is based on two key criteria:

1. **New Well Test Availability:** The system monitors the date of the last well test. Whenever new well test data becomes available beyond this date, automatic retraining is triggered to integrate the latest information.
2. **Parameter Changes:** If there is any addition or removal of parameters from the *Well_parameters_list*, the model is retrained to reflect these updates.

This approach guarantees that well models remain both current and optimally configured, automatically adapting to changes in data availability while maintaining rigorous quality controls. To further ensure data integrity and model reliability, the following rules are enforced:

- A minimum of three well tests per well is required to train a model.
- Any well test missing a selected parameter is discarded from the training process.
- Newly introduced parameter tags without historical data are excluded from the dataset.

Model Inference. The inference stage of the model provides flow rate estimates in two distinct ways. First, the inference pipeline is executed at regular 30-minute intervals. During each cycle, the most recent well parameter values obtained from the sensors are retrieved and aligned with the input requirements of each well. The trained machine learning models are then applied to infer the corresponding VFM estimates, which are subsequently made available for further analysis and operational use.

Second, the model is employed to conduct a sensitivity analysis on the selected well parameters used in the VFM flow rate estimation. For each parameter, an ensemble of values is generated from the sensor readings by defining an absolute range and appropriate sampling increments. A table of all parameter combinations is then constructed, and the VFM model is executed for each case to produce new flow rate estimates. This procedure enables the algorithm to determine the well conditions that most closely satisfy the defined production target.

In summary, optimization algorithms work by systematically adjusting input variables with aims to maximize predicted flow rates while strictly adhering to operational constraints. The algorithms iterate through hundreds of input combinations, running predictions to evaluate how each change impacts production outputs.

As field conditions change, the optimization process is repeated in real-time allowing the system to adapt continuously and ensuring that the operating strategy remains aligned with production objectives. By comparing predicted results against predefined production targets, a VFM-enabled system can identify and recommend in an automated way the optimized operational guidelines required to achieve the defined

production goals. These insights are valuable not only for maximizing oil recovery but also for maintaining long-term reservoir health and compliance with asset management requirements.

Contextual Awareness

A key innovation of this approach lies in the integration of contextual awareness, which uses an artificial intelligence (AI) engine to combine real-time data with operational knowledge in a way that simplifies decision-making. A large language model (LLM) plays a central role in this method by processing historical operational data, including intervention records, shutdown reports, and production logs, and creating a proximity vector that organizes the information and adds meaningful context to it. It then combines this output with real-time field measurements and compares them against their optimized operating envelopes. The result is a clear, human-readable recommendation presented in natural language (see Figure 3) that allows operators to easily understand and act on the information provided.



Figure 3—Workflow illustrating the integration of multiple data sources into a large language model (LLM) to generate context-aware operational recommendations.

By including past events in its data-crunching process, the LLM introduces a valuable layer of contextual awareness to its recommendations. This enables the system to enhance operating envelopes with insights that are aligned with the historical conditions of the asset, all the while ensuring that prior operational knowledge is not lost nor overlooked. This proactive approach helps operators address potential issues before they escalate and ensures that decisions are grounded in accurate technical knowledge.

At the same time, the LLM remains highly responsive to current field conditions, using real-time measurements to provide recommendations that reflect the current state of wells. Blending these two perspectives together, historical and real-time, creates a robust framework for decision-making that allows operators to receive insights that are both precise and immediately actionable. This minimizes reliance on trial-and-error methods, saving time and reducing inefficiencies in day-to-day operations.

Over the long term, the ability to learn from past events while staying responsive to real-time changes supports continuous improvement and ensures that production processes adapt to evolving reservoir conditions without losing the valuable lessons embedded in historical data.

Framework

A three-step approach is designed to tackle the structural challenges created by disconnected systems, fragmented workflows, and lack of standardization across data sources. In many cases, these limitations are the main factor preventing organizations from fully leveraging the wealth of information generated by their operations.

Step 1: Integration. The process begins by aggregating and organizing data from diverse sources into a single and unified digital platform. This includes consolidating input from real-time data historians (e.g.: PI System), production accounting systems (e.g.: Energy Components), nodal analysis tools (e.g.: Prosper and GAP), and internal operational logs (e.g., reports and logbooks). To facilitate this integration, data engineering pipelines are implemented to automatically extract, transform, and load the data, ensuring its standardization and complete alignment with the predefined data model (see Figure 4):

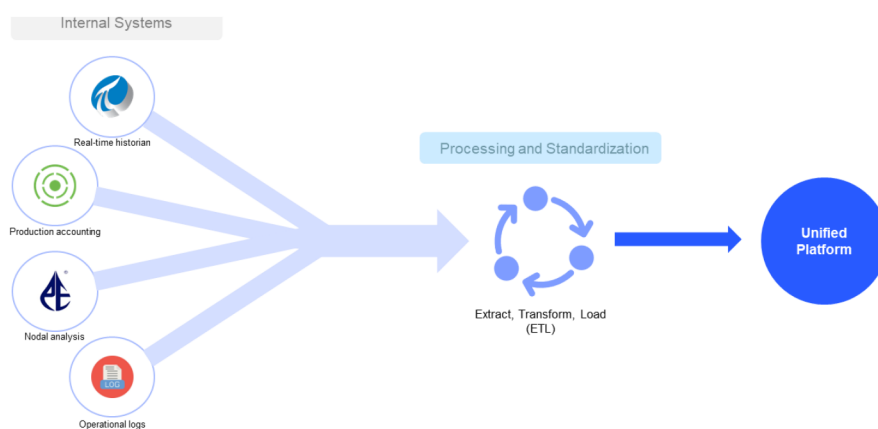


Figure 4—Data integration workflow illustrating the consolidation of inputs from internal systems into a unified platform via ETL processes and standardized data model.

By establishing a centralized and structured repository, this step ensures a consistent and accurate view of information, enabling a comprehensive understanding of well production systems and supporting any subsequent analytical process.

Step 2: Analysis. Once integrated, the data undergoes further processing through advanced analytics and machine learning techniques. In this stage, VFM models are trained and calibrated to generate dynamic operating envelopes. These envelopes are subsequently reincorporated into the unified platform to support real-time monitoring of live field data. If deviations are detected, the system triggers an automated workflow to gather contextual information about the anomaly. This includes details about the well, operating conditions, and any relevant historical data. By evaluating these inputs, the system identifies the likely root cause of the problem enabling it to suggest precise and effective corrective actions tailored to the specific issue.

Step 3: Recommendation. The final step converts analytical insights into actionable recommendations and presents them through an easy-to-use interface. When a deviation is identified, users receive instantaneous notifications providing initial information about the anomaly (see Figure 5). This includes details like the concerned well, the specific parameter requiring attention, the current field value, and its corresponding operating envelope:

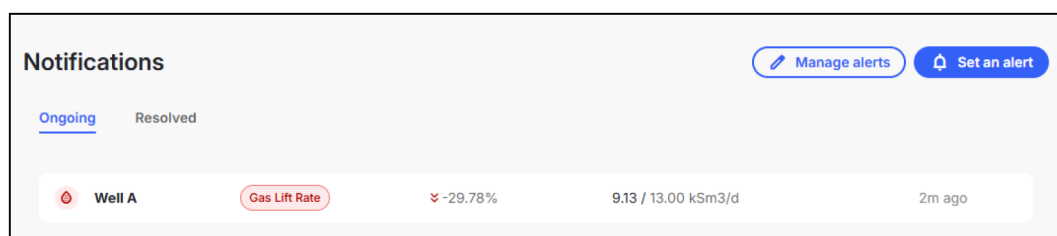


Figure 5—Real-time alert interface showing a significant drop in gas lift rate for Well A, with current and target values displayed.

Recommendations are provided in clear natural language to ensure operators can easily interpret and act upon them without ambiguity (see Figure 6). Suggestions might involve adjusting the choke opening to optimize WHP and stabilize production rates or recalibrating gas lift rates (GLR) and electric submersible pump (ESP) setpoints to improve artificial lift efficiency. In all cases, these suggestions are grounded on current field measurements and historical knowledge to address deviations and maintain well performance within operational targets:

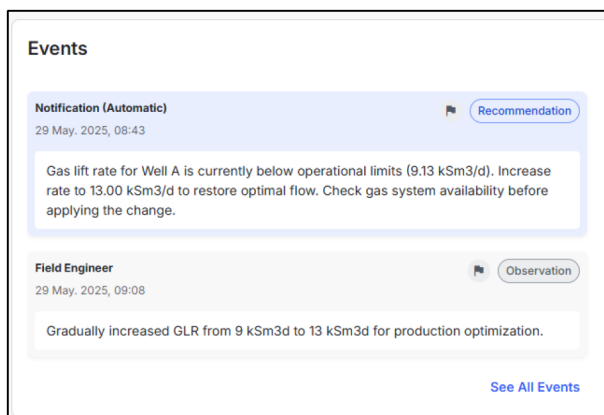


Figure 6—Event log showing automated system recommendation in natural language and field engineer response.

To aid in decision-making, the interface also provides intuitive visual tools designed to streamline the analysis of operational data (see Figure 7). Trend charts allow users to monitor parameter fluctuations over time, offering insights into the progression of anomalous behavior or overall performance indicators. A traffic-light system complements this by categorizing conditions into green, yellow, or red states, quickly signaling urgency and helping users prioritize corrective actions based on operational envelopes:

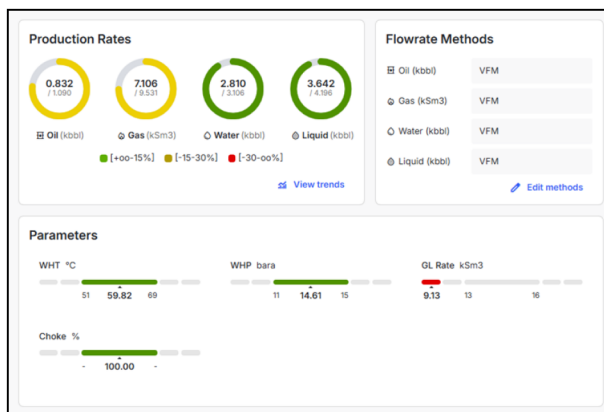


Figure 7—Performance dashboard displaying real-time production rates and key operating parameters, highlighting deviations in gas lift rate relative to optimal operating envelopes.

Beyond improving operational efficiency, this step fosters teamwork by sharing a centralized and transparent view of field performance with cross-disciplinary teams. Engineers, technologists, and operators can collaborate using real-time data to track metrics, identify anomalies, and analyze system behavior. By drawing on diverse expertise and aligning goals through unified information, teams can make better-informed decisions, enhance problem-solving approaches, and respond to challenges more promptly and effectively.

Results

This approach was implemented in two mature offshore assets operated by TotalEnergies, one in the Gulf of Guinea (Africa) and the other in the Persian Gulf (Middle East), both characterized by aging infrastructure and growing operational constraints. In the Gulf of Guinea, gas lift-activated wells operating at sub-optimal envelopes provided multiple areas for optimization. Improvements to choke control ensured downhole flow remained within erosion limits, preserving completion integrity and yielding a 15% production increase. Choke malfunctions caused by excessive production rates were also identified and rectified, restoring stable

flow and contributing with an extra 7% gain. Low WHP and WHT, indicative of liquid buildup in the tubing, were addressed by parameter adjustments, adding a 1% increase. During a 5-bar rise in the production header pressure, dynamic choke adjustments minimized backpressure, maintaining flow rates stable and securing a 3% production improvement. Reservoir monitoring was improved through the use of VFMs, allowing for more precise adjustments to voidage replacement strategies and improving pressure maintenance across the reservoir.

In the Persian Gulf, where wells used ESPs, optimization focused on pump performance and zonal flow adjustments. Increasing motor frequency within its operating limits enabled higher production rates while maintaining ESP reliability, contributing to a 3% production increase. Similarly, variations in the WHT signalled a gradual ingress of water into the tubing, leading operators to shut down water-prone zones using surface-controlled actuators, resulting in a 7% production gain.

Across both contexts, the methodology provided concrete results and delivered a cumulative production gain of over 150,000 barrels by resolving bottlenecks and enhancing operational control. At the same time, production efficiency was significantly improved in both regions with optimized equipment use and enhanced well management. The results highlight the wide-ranging applicability of this methodology, which balances technical constraints and field-specific requirements, offering a scalable approach for mature offshore assets worldwide.

Conclusion

The presented work demonstrates how the digital transformation in the Oil & Gas industry can improve operational efficiency when advanced data-driven solutions are appropriately implemented. Despite the typical challenges associated with legacy systems, data silos, and inconsistent workflows, the methodology outlined here successfully integrated multi-source data to capture complex system dynamics and enable prescriptive guidance to operators. By applying the concept of dynamic operating envelopes and enriching them with contextual knowledge, operators were able to reduce their dependency on manual analysis and optimize the production performance of wells. The results from field trials in both the Gulf of Guinea and the Persian Gulf further confirm the broad applicability of this approach, where the unified platform acted as a foundational tool to transform production insights into real field actions.

While this technique offers significant advantages in optimizing well operations and boosting production efficiency, it is not without limitations. A significant challenge is its dependence on accurate, consistent, and comprehensive data from multiple sources, with inaccuracies leading to reduced model performance and consequently suboptimal recommendations. Integrating various data systems often requires substantial investment in infrastructure, software, and skilled personnel, which may pose challenges for organizations with limited resources. Machine learning models also require continuous calibration and maintenance to adapt to ensure their reliability over time, and the workforce that uses these tools needs to be able to interpret and act on the technical insights provided by them. These limitations suggest that while promising, the approach depends on a supportive ecosystem of reliable infrastructure, skilled professionals, and continuous improvements to maximize its potential.

Beyond the immediate production gains, this approach provides a clear pathway for systems that leverage AI-driven insights to enable broader optimization at field and asset levels. These advancements not only improve efficiency but also foster sustainable and resilient operations, positioning the industry for continued success in an increasingly data-centric global landscape.

Nomenclature

- AI - Artificial Intelligence
- Choke - Flow Control Valve
- ETL - Extract, Transform, Load

ESP - Electric Submersible Pump
 GLR - Gas Lift Rate
 LLM - Large Language Model
 ML - Machine Learning
 PI - Plant Information (System)
 VFM - Virtual Flow Meter
 WHP - Wellhead Pressure
 WHT - Wellhead Temperature
 WOE - Well Operating Envelope

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